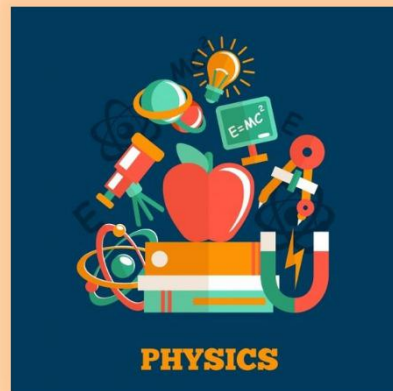




1.1 Physics Understanding

1.2 Scalars and Vectors

(Laboratory works)

1.3 Physical Measurement,
Plotting and
Interpreting Linear
Graph





Learning Outcomes

1.1 Physics Understanding

- a) **State** basic quantities and their respective SI units: length (m), time (s), mass (kg), electrical current (A), temperature (K), amount of substance (mol) and luminosity (cd).
- b) **State** derived quantities (in terms of basic quantities) and their respective units and symbols: velocity (ms^{-1}), acceleration (ms^{-2}), *work* (J), force (N), pressure (Pa), energy (J), power (W) and frequency (Hz).
- c) **Perform** conversion between SI units.



Learning Outcomes

1.2 Scalars and Vectors

- a) **Define** scalar and vector quantities.
- b) **Compare** scalar and vector quantities.
- c) **Resolve** vector into two perpendicular components (x and y axes).
- d) **Determine** resultant vector of two vector component.
- e) Write a laboratory report
(Experiment 1: Physical Measurement)
(Experiment 2: Plotting and interpreting linear graph)

1.1 (a) Physics Understanding

Physical quantities is a quantity that can be measured.

Physical quantities can be derived to:

- Base quantities
- Derived quantities

Base quantity is defined as **quantity which cannot be derived from any physical quantities.**

Table 1 shows basic quantities.

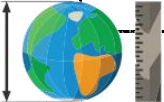
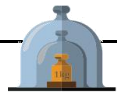
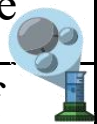
Basic Quantity	SI unit / symbol	Symbol of dimension
 Length	meter / m	L
 Mass	kilogram / kg	M
 Time	second / s	T
 Electric current	ampere / A	A
 Temperature	kelvin / K	θ
 Amount of substance, n	mole / mol	N
 Luminous intensity	candela / cd	J

Table 1

1.1 (b) Physics Understanding

Physical Quantities	Equation	SI unit
Velocity	$v = \frac{s}{t}$	m s^{-1}
Acceleration	$a = \frac{v}{t}$	m s^{-2}
Momentum	$p = mv$	kg m s^{-1}
Work	$W = Fs$	$\text{kg m}^2 \text{s}^{-2}$ (J)
Force	$F = ma$	kg m s^{-2} (N)
Pressure	$p = \frac{F}{A}$	$\text{kg m}^{-1} \text{s}^{-2}$ (Pa)
Energy	$E = mgh$	$\text{kg m}^2 \text{s}^{-2}$ (J)
Power	$P = \frac{W}{t}$	$\text{kg m}^2 \text{s}^{-3}$ (W)
Frequency	$f = \frac{1}{\text{period}}$	s^{-1} (Hz)

Table 2

Table 2 show derived quantities and their units in terms of base units.

1.1 (c) Physics Understanding

- Sometimes it is necessary to work with measurements that are much smaller or much larger than the standard SI units.
- In these situations, we can use other units that are related to the standard SI units by a multiple of ten. Prefixes are used to denote the different powers of ten.
- For example, the prefix 'kilo' means 1000, or 10^3 , while the prefix 'micro' means 0.000001, or 10^{-6} .
- The list for prefixes can be seen in Table 3.
- These prefixes can be applied to any SI unit: for example, 0.001 second is 1 millisecond (ms).

Prefix	Multiple	Abbreviation
tera	10^{12}	T
giga	10^9	G
mega	10^6	M
kilo	10^3	k
desi	10^{-1}	d
centi	10^{-2}	c
mili	10^{-3}	m
micro	10^{-6}	μ
nano	10^{-9}	n
pico	10^{-12}	p

Table 3

1.1 (c) Physics Understanding

- Because different systems of units are in used, it is important to know how to convert from one unit to another unit.
- When physical quantities are added, subtracted, multiplied or divided in an algebraic equation, the unit can be treated like any other algebraic quantity.

Example 1

Suppose you want to convert 1 km to meter:

$$1\cancel{\text{km}} \times \frac{10^3 m}{1\cancel{\text{km}}}$$

$$= 1 \times 10^3 m$$

Exercise 1

Try and convert the following quantities into the units as stated:

- a) 1 pF to F
- b) 4 km to m
- c) 67mV to V
- d) 0.7 dm to m
- e) 0.098 GW to W
- f) 13 mA to A

(Answers: $1 \times 10^{-12} F$,
 $4 \times 10^3 m$,
 $67 \times 10^{-3} V$,
 $7.0 \times 10^{-2} m$,
 $9.8 \times 10^7 W$,
 $1.30 \times 10^{-2} A$)

Example 2

Suppose you want to convert 30 m to km:

$$\begin{aligned} & \cancel{30\cancel{m}} \times \frac{1\cancel{km}}{10^3\cancel{m}} \\ &= \frac{30 \times 1\cancel{km}}{10^3} \\ &= 3.00 \times 10^{-2}\cancel{km} \end{aligned}$$

Exercise 2

Try and convert the following quantities into the units as stated:

- a) 20 m to cm
- b) 87 A to kA
- c) 190 Ω to k Ω
- d) 0.005 F to pF
- e) 56 Hz to MHz

(Answers: $2.00 \times 10^3 m$,
 $8.70 \times 10^{-2} kA$,
 $1.90 \times 10^{-1} k\Omega$,
 $5.00 \times 10^9 pF$,
 $5.60 \times 10^{-5} MHz$)

Example 3

Suppose you want to convert 43 mm^3 to m^3 :

$$\begin{aligned} & 43 \text{ mm}^3 \times \left(\frac{10^{-3} \text{ m}}{1 \text{ mm}} \right)^3 \\ &= 43 \cancel{\text{mm}}^3 \times \left(\frac{10^{-9} \cancel{\text{m}^3}}{1 \cancel{\text{mm}^3}} \right) \\ &= \frac{43 \times 10^{-9} \text{ m}^3}{1} \\ &= 4.30 \times 10^{-8} \text{ m}^3 \end{aligned}$$

Exercise 3

Try and convert the following quantities into the units as stated:

- a) 19 km^2 to m^2
- b) 9 m^2 to cm^2
- c) 0.03 m^2 to mm^2
- d) 36 cm^3 to m^3
- e) 24 mm^2 to m^2
- f) 4 m^2 to dm^2
- g) 608 m^3 to cm^3

(Answers: $1.90 \times 10^7 \text{ m}^2$,
 $9.00 \times 10^4 \text{ cm}^2$,
 $3.00 \times 10^4 \text{ mm}^2$,
 $3.60 \times 10^{-5} \text{ m}^3$,
 $2.40 \times 10^{-5} \text{ m}^2$,
 $4.00 \times 10^2 \text{ dm}^2$,
 $6.08 \times 10^8 \text{ cm}^3$)

Example 4

Suppose you want to convert 50 kmh^{-1} to ms^{-1} :

$$\begin{aligned} & \frac{50 \cancel{\text{km}}}{1 \cancel{\text{h}}} \times \frac{1 \cancel{\text{h}}}{(60 \times 60) \text{s}} \times \frac{10^3 \text{m}}{1 \cancel{\text{km}}} \\ &= \frac{50 \times 1 \times 10^3 \text{m}}{1 \times (60 \times 60) \text{s} \times 1} \\ &= 13.89 \text{ms}^{-1} \end{aligned}$$

Exercise 4

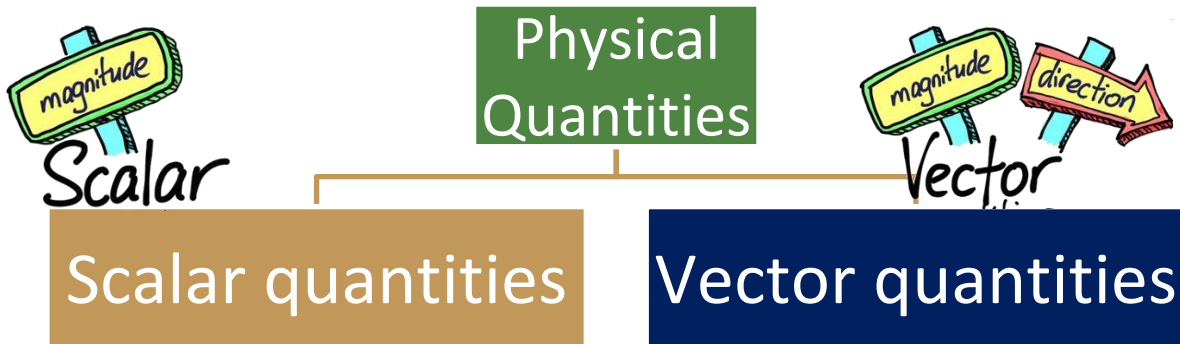
Try and convert the following quantities into the units as stated:

- a) 560 km h^{-1} to m s^{-1}
- b) 67 m s^{-1} to km h^{-1}
- c) 0.089 km h^{-2} to m s^{-2}
- d) 56 g cm^{-3} to kg m^{-3}
- e) 78 kg m^{-3} to g cm^{-3}
- f) 45 N m h^{-1} to N m s^{-1}
- g) $56 \text{ cm}^3 \text{ min}^{-1}$ to $\text{m}^3 \text{ s}^{-1}$

(Answers:

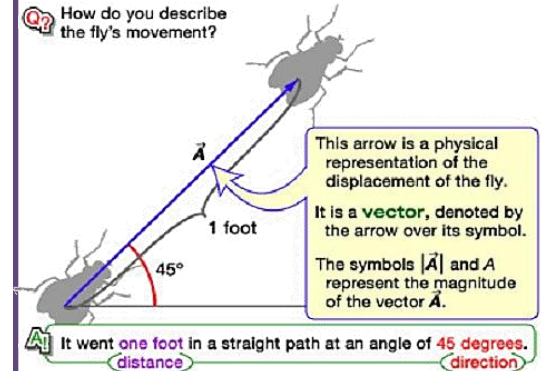
$1.56 \times 10^2 \text{ m s}^{-1}$,
 $2.41 \times 10^2 \text{ km h}^{-1}$,
 $6.87 \times 10^{-6} \text{ m s}^{-2}$,
 $5.60 \times 10^4 \text{ kg m}^{-3}$,
 $7.80 \times 10^{-2} \text{ g cm}^{-3}$,
 $1.25 \times 10^{-2} \text{ N m s}^{-1}$,
 $9.33 \times 10^{-7} \text{ m}^3 \text{ s}^{-1}$)

1.2 Scalars and vectors



- Is defined as quantity with **magnitude** only.
- Example: mass, distance, speed, work, pressure, current, temperature.
- Is defined as quantity with **both magnitude and direction**.
- Example: displacement, velocity, force, momentum, impulse, torque, electric field, magnetic field.

- ✓ Vectors are typically illustrated by drawing an ARROW above the symbol e.g. $\vec{x}$, $\vec{v}$, $\vec{a}$, $\vec{F}$
- ✓ Vector can be represented graphically using arrow.

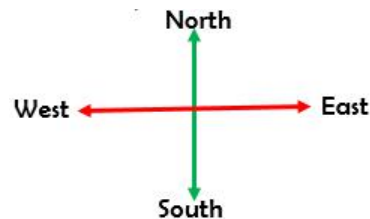


- ✓ When doing any mathematical operation on a vector quantity (like adding, subtracting, multiplying ..) we have to consider both the magnitude and the direction.

Direction of Vectors

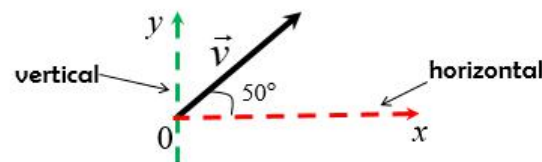


- Can be represented by using:
 - a) **Direction of compass**, i.e east, west, north, south, north-east, north-west, south-east and south-west



- b) **Angle with a reference line**

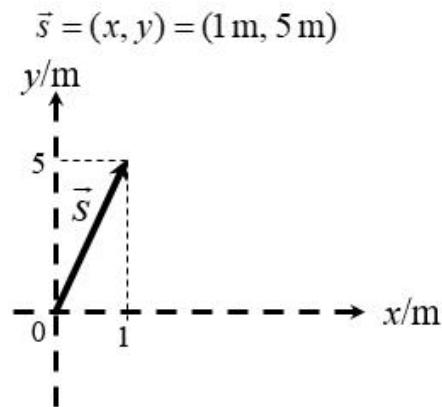
e.g. A boy throws a stone at a velocity of 20 m s^{-1} , 50° above horizontal.



Direction of Vectors



- Can be represented by using:
 - c) **Cartesian** coordinates
 - 2-Dimension (x and y axes)

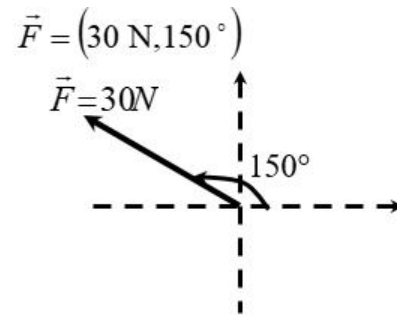


Direction of Vectors

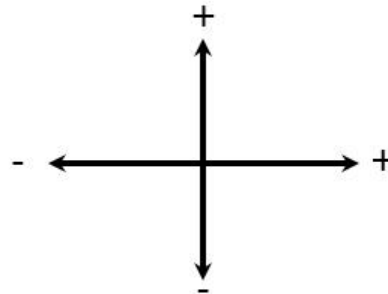


- Can be represented by using:

d) **Polar** coordinates



e) **Denotes** with **+** or **-** signs.

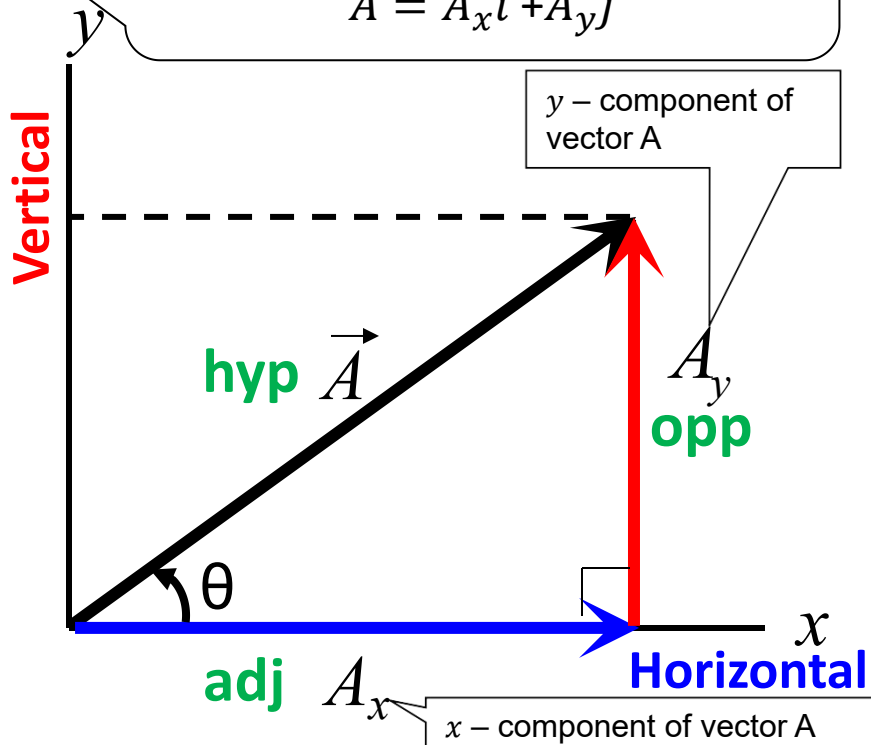


Resolving a vector into 2 perpendicular components



A vector may be expressed in terms of its components.

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$



with the aid of trigonometry:

$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{A_x}{\vec{A}} \mapsto A_x = \vec{A} \cos \theta$$

For the comp which is **adjacent** to angle θ , we use **cos**

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{A_y}{\vec{A}} \mapsto A_y = \vec{A} \sin \theta$$

For the comp which is **opposite** to angle θ , we use **sin**

Magnitude of vector A :

$$|A| = \sqrt{A_x^2 + A_y^2}$$

Direction of vector A :

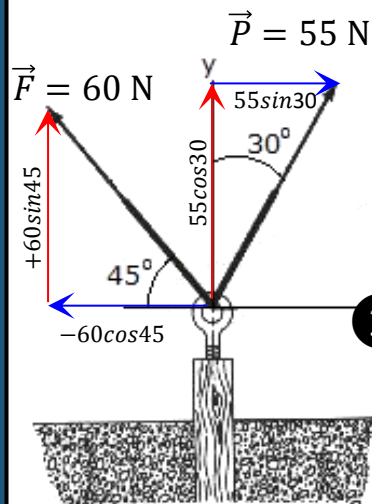
$$\theta = \tan^{-1} \left| \frac{A_y}{A_x} \right|$$

* θ is measured relative to x axis.

VECTOR ANALYSIS : Determine resultant vector



Flow map : Problem solving strategy
(using components to add vectors and determine Resultant vector)



1 Resolve each vector(s) into its x & y component

Denotes with + or - signs.



2 Add comp. x and y separately

STRICTLY DO NOT add x components to y components

3 Visualize the total component for x and y, resultant vector and the direction.

4 Calculate the magnitude of resultant vector :

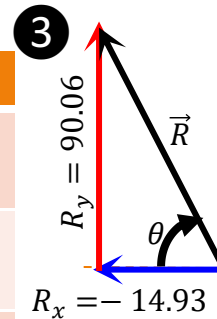
$$|R| = \sqrt{R_x^2 + R_y^2}$$

5 Calculate the direction of resultant vector :

$$\theta = \tan^{-1} \left| \frac{R_y}{R_x} \right|$$

θ is measured relative to x axis.

	Comp x	Comp y
1 $\vec{F}$	$-60\cos 45$ -42.43 N	$60\sin 45$ 42.43 N
$\vec{P}$	$55\sin 30$ 27.5	$55\cos 30$ 47.63 N
2 $\vec{R}$	-14.93 N	90.06 N



4

$$|R| = \sqrt{R_x^2 + R_y^2}$$

$$= \sqrt{(-14.93)^2 + (90.06)^2} = \tan^{-1} \left| \frac{90.06}{-14.93} \right|$$

$$|R| = 91.29 \text{ N}$$

5

$$\theta = \tan^{-1} \left| \frac{R_y}{R_x} \right|$$

$$\theta = 80.6^\circ \text{ above } -x$$

Exercise 5

1. Two displacement vector **A** and **B** are shown in **FIGURE 5.1**. Find the magnitude and direction of the resultant displacement.

(Answer: 1.68 km, 7.5° below negative x- axis)

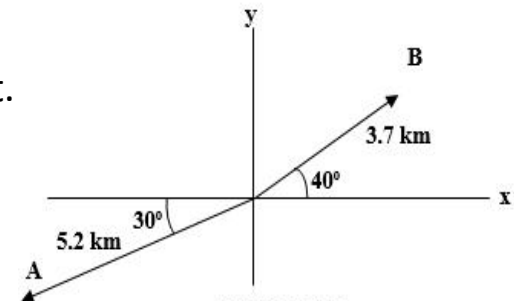


FIGURE 5.1

2. Two forces $F_1 = 8\text{ N}$ and $F_2 = 12\text{ N}$ are acting on a wooden block shown in **FIGURE 5.2**. Calculate the magnitude and direction of the resultant force acting on the wooden block.

(Answer: 19.35 N, 11.93° above positive x- axis)

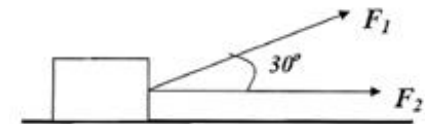


FIGURE 5.2

See you in the next chapter!

Chapter 2: Kinematics of Linear Motion

